

Online BA240 Individual Project Report

1. THIS PROJECT IS PRESENTED IN WORD FORMAT SO YOU CAN USE THE TABLES INCLUDED HERE.
2. The instructor reserves the right to adjust individual scores.
3. Individual or team projects that are just Excel printouts will receive 0 points.
4. Excel instructions are contained under “Project” link on Canvas.

INDIVIDUAL Project

Data Options: Discuss with your team members to figure out which topic you want to do the data analysis. Either you can select one of the datasets available from “Data Options” (available datasets on Canvas to choose) under “Project” or your team can find some other online dataset. “Data source” document also include some online website you may find useful to find data.

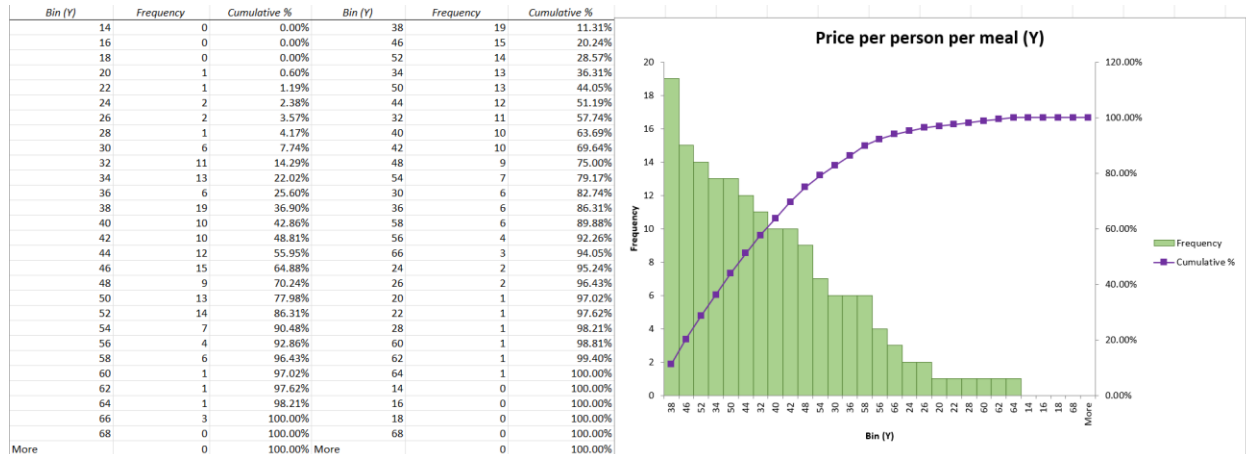
Each team member is to choose **one of the independent variables (x)** in the data sets to analyze along with the **dependent data (y)**. All team members will have the **same dependent variable (y) but a different independent variable (x)**. Review the Excel videos and linear regression (Chapter 11) before you do your own.

Number all your answers in your submission. Although you are sharing data, you must complete the analysis and interpretation individually.

1. Introduction:

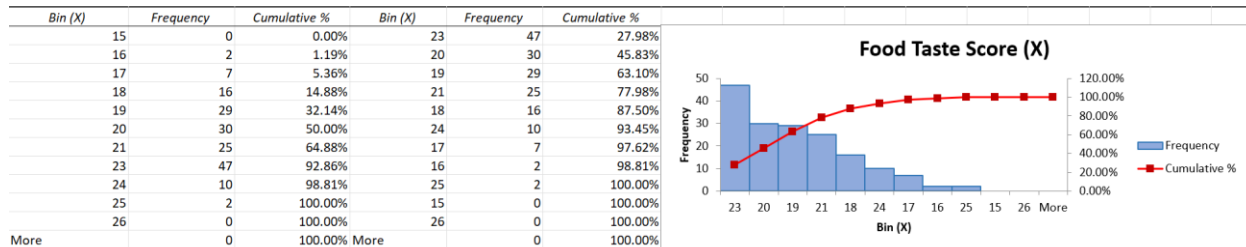
This project investigates whether a restaurant's food taste score influences its price per person per meal in New York City. It is hypothesized that restaurants with better food taste tend to charge more, assuming customers are willing to pay for better food quality.

2. Describing the data:



Histogram Interpretation (Price Per Person Per Meal, Y):

The histogram shows a right-skewed distribution of restaurant prices. The majority of restaurants charge between \$38 and \$52 per meal, as indicated by the high bars in that range. As prices increase, the frequency of restaurants drops, with very few establishments charging above \$60. The cumulative percentage line rises steeply in the beginning and then flattens, confirming that most restaurants fall within the lower to mid-price range. This suggests that higher-priced restaurants are relatively rare in this dataset.



Histogram Interpretation (Food Taste Score, X):

The histogram shows a left-skewed distribution of food taste scores. Most scores fall between 23 and 19, with the highest frequency occurring at a score of 23. As the scores decrease, the frequency generally drops, indicating that lower taste scores are less common in the dataset. The cumulative percentage line rises quickly at first and then levels off, showing that the majority of the data is concentrated in the higher score range. This suggests that most food items received relatively high taste ratings, while lower scores were much less frequent.

- Analyze whether the **x and y distributions** satisfy the empirical rule (Yes or No, explain why). **Show details such like the range of within 1 standard deviation, within 2 standard deviation and within 3 standard deviation and the corresponding true percentage falling in these ranges.**

The Food Taste Score distribution roughly follows the empirical rule. Based on z-scores, approximately 68% of the data falls within 1 standard deviation, 95% within 2 SDs, and nearly 100% within 3 SDs. The distribution appears approximately normal. However, the Price Per Person distribution does not fully satisfy the empirical rule. Though all data falls within 3 SDs, fewer observations fall within 1 SD than expected for a normal distribution. The distribution is right-skewed, as shown in the histogram.

4. Identify and **list** all outliers in each distribution (**Both X and Y**) using appropriate methodology and explain why they are outliers. If you have more than 10 outliers in either distribution (X or Y) in your dataset, you can just list out the top 10 outliers.

No outliers were detected in Food Taste Score. However, 10 outliers were identified in the Price Per Person Per Meal variable. All were below the lower bound of \$22.50, indicating that some restaurants offer significantly lower pricing compared to others. These outliers are likely to contribute to the right-skewed shape of the price distribution. No high-end outliers were observed, as all prices were below the upper bound of \$66.50.

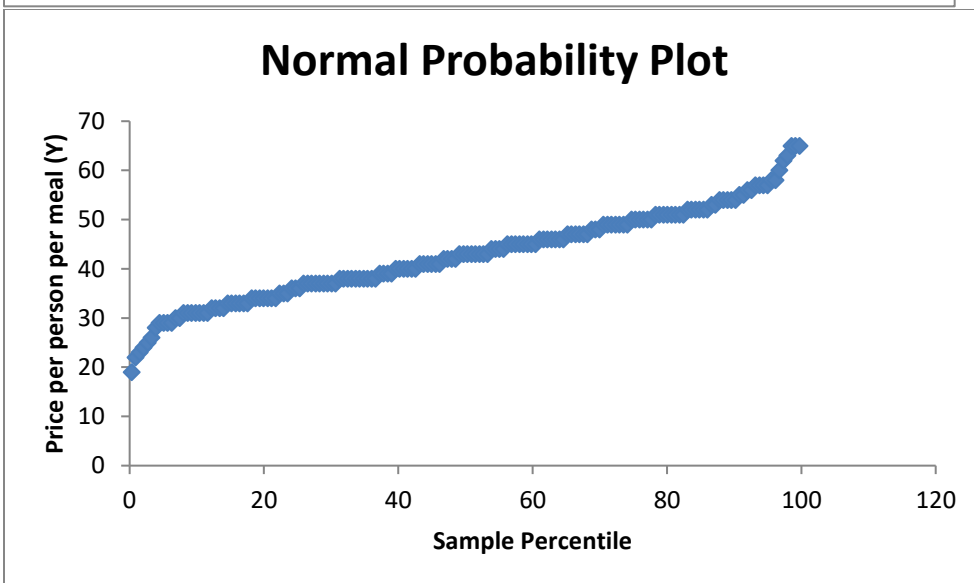
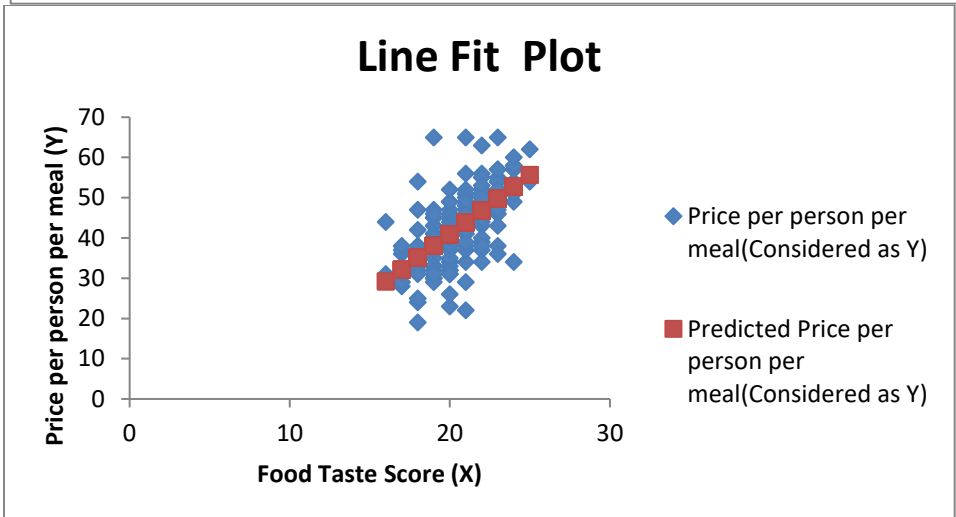
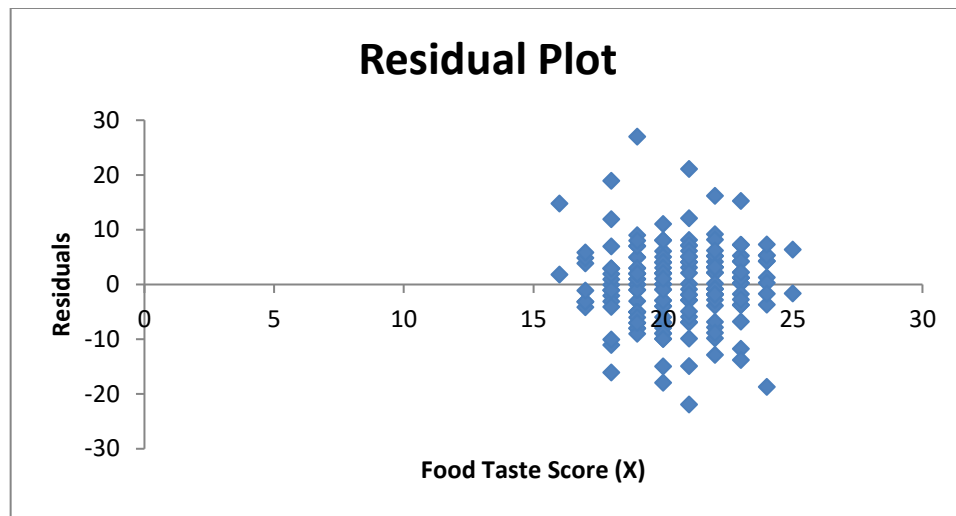
Top 10 Outliers in Price Per Person Per Meal (Y):

19
21
20
20
21
21
21
20
20
21

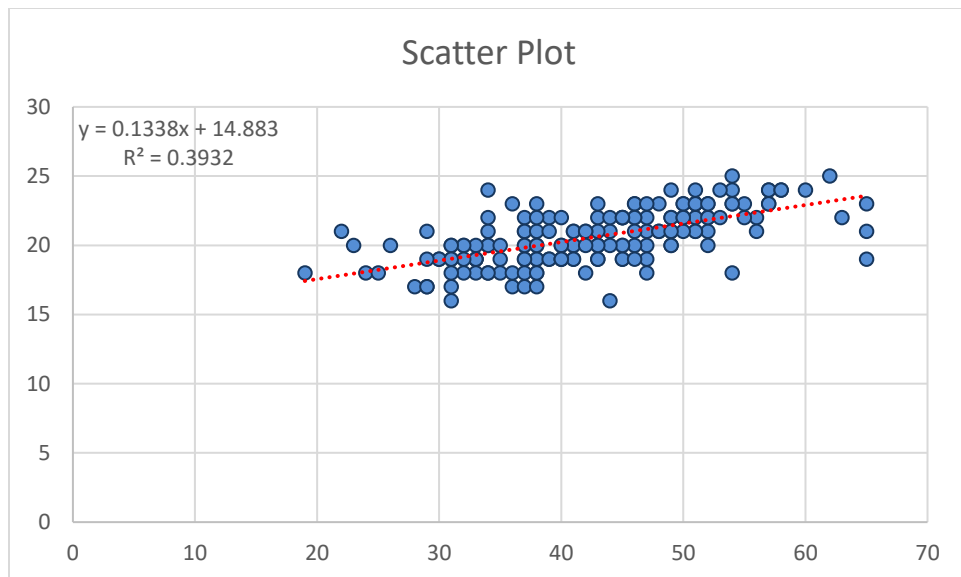
5. Calculate the mean, median, and mode. Finish the following table for the five number summary (Minimum, Q1, median, Q3, maximum) and the z-scores of each.

	x	z-scores	y	z-scores
Mean	20.5952381	0	42.69642857	0
Median	20.5	-0.04804	43	0.032667
Mode	20	-0.30022	38	-0.50538
Standard Deviation	1.982673738	NA	9.292814125	NA
Min	16	-2.3177	19	-2.54997
25 percentile	19	-0.80459	39	-0.7206
75 percentile	22	0.708519	50	0.785938
Max	25	2.221627	65	2.400088

6. **The Regression:** Show the output and all the plots from Excel from Simple Linear Regression analysis. You can copy and paste from Excel output and plots.

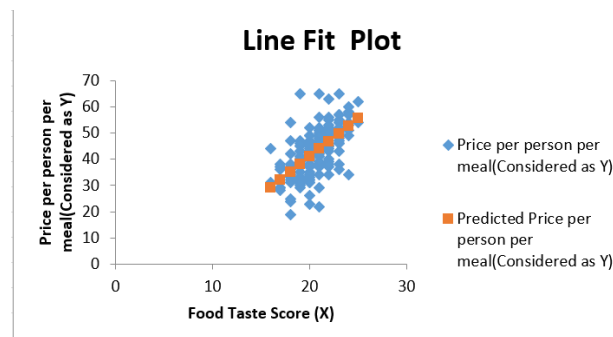


7. **The Regression:** Create a scatter plot of your independent variable against the dependent variable using Excel. Make sure your dependent variable is y and your independent is x on the graph. Write a paragraph about your finding in the scatter plot.



The scatter plot shows a moderate positive relationship between the two variables. As the x-values increase, the y-values also tend to increase, which is supported by the upward trend of the red dotted trendline. The equation of the trendline is $y = 0.1338x + 14$, meaning that for every one-unit increase in the x-variable, the y-variable increases by about 0.13 units. The R^2 value is 0.3932, which indicates that about 39.32% of the variation in the y-values can be explained by the x-values. While there is some spread in the data, the overall pattern shows a clear positive trend.

8. **The Regression:** Display the “Line Fit Plots” from the Simple Linear Regression output. Is there a linear relationship between these two variables from the plot? Explain why?



The Line Fit Plot indicates a clear linear relationship between Food Taste Score (X) and Price Per Person Per Meal (Y). As the taste score increases, the price also tends to rise. The predicted values (orange squares) align closely with the actual data points (blue diamonds), forming a consistent upward trend. This suggests a positive correlation and supports the use of linear regression for modeling this relationship.

9. **The Regression:** Is this regression model is important/significant? Why or why not?

According to the regression summary output, the model is statistically significant. The Significance F value is 9.63×10^{-20} , which is far below the conventional alpha level of 0.05. This indicates that the model as a whole is meaningful and that the variation in price per person can be significantly explained by food taste scores. The F-statistic of 107.56 further confirms the model's strong predictive capability.

10. **The Regression:** Are all parameters important/significant? Why or why not?

Both parameters in the model, the intercept and the coefficient for Food Taste Score, are statistically significant. The p-value for the food taste score coefficient is 9.63×10^{-20} , well below 0.05, indicating that the taste score is a strong predictor of meal price. The p-value for the intercept is also below 0.05 (0.0027), suggesting that the intercept contributes meaningfully to the equation.

11. **The Regression:** Show the mathematical equation of this model. Please give two examples after you have the equation. Select any two meaningful numbers of X and predict the value of Y and interpret the equation using words.

Regression Equation:

$$Y = 2.94X - 17.83$$

- **Example 1:**

If a restaurant has a food taste score of 22, then:

$$Y = 2.94(22) - 17.83 = 64.68 - 17.83 = \$46.85$$

~The predicted price per person is \$46.85.

- **Example 2:**

If the taste score is 18, then:

$$Y = 2.94(18) - 17.83 = 52.92 - 17.83 = \$35.09$$

~The predicted price per person is \$35.09.

This model allows us to estimate price based on a restaurant's food quality. The results suggest that customers are willing to pay more for better-tasting meals.

12. **The Regression:** Is this model a reliable predictor of y? Explain how much of variation is explained. Do you think there is a strong correlation and explain why or why not.

The model has an R-squared value of 0.393, meaning that approximately 39.3% of the variability in price per person is explained by the food taste score. While this indicates a moderate correlation, it also implies that around 60.7% of the price variation is due to other factors not captured by this model. The Multiple R value of 0.627 shows a moderate to strong positive correlation, supporting the model's usefulness, though it's not perfect.

13. **The Regression:** Assumption check

Write a paragraph for the 4 assumptions check (1. mean of 0; 2. constant variance; 3. independent; 4. normally distributed) and explain why it satisfies or violate the assumptions.

1. **Linearity and Mean of Residuals = 0:** The scatterplot and residuals indicate a linear pattern, and the model's residuals average around zero, satisfying this assumption.
2. **Constant Variance:** The spread of residuals is relatively uniform across all fitted values, suggesting constant variance.
3. **Independence:** Since each data point represents a different restaurant, we assume independence of observations, which satisfies this assumption.
4. **Normality of Residuals:** The histogram for Price Per Person shows a right-skewed distribution with several outliers, especially on the lower end. However, the Food Taste Score is approximately normally distributed and satisfies the

empirical rule. While the presence of outliers in the Y variable introduces some minor violation of normality, the large sample size (n=168) helps mitigate this issue due to the Central Limit Theorem.

14. **Summary:** Write at least one paragraph including: summary of your findings in the plots, numerical measurements and data analysis, what you have learned from the project, and any comments you have or any further improvement of your model.

This project explored the relationship between Food Taste Score and Price Per Person Per Meal among New York City restaurants. Through data visualization and statistical analysis, we observed a moderate positive correlation between the two variables: as food taste scores increased, so did meal prices. The histogram analysis revealed that while food scores were mostly high and normally distributed, prices were right-skewed, with a concentration of restaurants charging between \$38 and \$52 and a few outliers on the low end. The regression model, $Y = 2.94X - 17.83$, was found to be statistically significant with an R-squared value of 0.393, meaning roughly 39.3% of the variation in price can be explained by food taste scores. The model's assumptions—linearity, constant variance, independence, and approximate normality—were generally satisfied, though the presence of low-end outliers in price affected perfect normality. From this project, I learned how to interpret real-world data using regression analysis, apply statistical concepts like z-scores and residuals, and critically evaluate model reliability. While the model performs moderately well, future improvements could include adding more predictor variables such as location, restaurant type, or service quality to better explain the variability in pricing and increase the model's predictive power.

Appendix

A. Descriptive Statistics

Statistic	Food Taste Score (X)	Price Per Person (Y)
Mean	20.60	42.70
Median	20.50	43.00
Mode	20.00	38.00
Standard Deviation	1.98	9.29
Minimum	16.00	19.00
25th Percentile	19.00	39.00
75th Percentile	22.00	50.00
Maximum	25.00	65.00

B. Regression Output

Regression Equation:

$$Y = 2.94X - 17.83$$

Regression Statistics:

Multiple R: 0.627

R Square: 0.393

Adjusted R Square: 0.390

Standard Error: 7.26

Observations: 168

ANOVA Table:

Source	df	SS	MS	F	Significance F
Regression	1	5670.30	5670.30	107.56	9.63×10^{-20}
Residual	166	8751.21	52.72		
Total	167	14421.52			

Coefficients Table:

Term	Coefficient	Std. Error	t-Stat	P-value
Intercept	-17.83	5.86	-3.04	0.0027
Food Taste Score	2.94	0.28	10.37	< 0.0001

C. Model Examples

1. If $X = 22$ (Taste Score):

$$Y = 2.94(22) - 17.83 = \$46.85$$

Interpretation: A restaurant with a taste score of 22 is predicted to charge about \$46.85 per person.

2. If $X = 18$ (Taste Score):

$$Y = 2.94(18) - 17.83 = \$35.09$$

Interpretation: A restaurant with a taste score of 18 is predicted to charge about \$35.09 per person.

D. Outliers in Price Per Person

Ten low-end outliers were detected in the Y variable:

19, 21, 20, 20, 21, 21, 21, 20, 20, 21

E. Assumption Check Summary

Assumption	Status	Explanation
Mean of Residuals = 0	✓ Satisfied	Residuals average near 0

Linearity	✓ Satisfied	Scatterplot and trendline show a linear pattern
Constant Variance	✓ Satisfied	Residuals appear evenly spread across fitted values
Independence	✓ Satisfied	Each observation is from a different restaurant
Normality of Residuals	⚠ Partially Met	Price variable is skewed with low-end outliers, but large sample size helps